Gookho Song

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Research Interests

Core fields: metasurfaces · wavefront shaping · computational optics · physics-guided deep learning · holographic imaging · Fourier ptychography · computational spectroscopy & hyperspectral imaging.
I build optical encoders (disordered and inverse-designed metasurfaces, scattering media) paired with physics-informed reconstruction to enable compact, snapshot multimodal imaging (spectrum, phase and depth) that conventional optics cannot achieve.

Education

Korea Advanced Institute of Science and Technology (KAIST) 2022–present
Ph.D. Candidate, Bio and Brain Engineering Daejeon, Korea

- **Advisor:** Prof. Mooseok Jang
- **Focus:** metasurfaces; multimodal (hyperspectral, phase, depth) imaging; random/computational encoding

Korea Advanced Institute of Science and Technology (KAIST) 2020–2022
M.S., Bio and Brain Engineering · GPA 4.04 / 4.3 Daejeon, Korea

- **Advisor:** Prof. Mooseok Jang
- **Thesis:** *Differential detection of ultrasound-modulated light for deep tissue imaging*

Inha University 2016–2020
B.S., Electronic Engineering, double major in Biotechnology · GPA 4.11 / 4.5 Incheon, Korea

Honors & Awards

Student Paper Prize — Optica Imaging Congress, Computational Optical Sensing and Imaging (COSI) · Seattle, WA, USA Aug 2025

- Sole winner in the COSI track, awarded for an oral presentation.

Best Student Paper Award, 1st Place — Next-Generation Lithography & Patterning Conference · Korea 2024

- 1st place; research grant of ₩1,000,000. Topic: metasurface wavefront sensor for semiconductor metrology.

Best Presentation Award — KAIST Optics Group Student Poster Competition Jul 2024

- Topic: compact spectrometer based on a double-layer disordered metasurface.

Journal Articles

G. Song = applicant · † equal contribution · * corresponding author
Google Scholar (Sep 2026): 231 citations · h-index 8 · i10-index 7

1. Y. Kim†, **G. Song**†, H. Kim, C. Lee, M. Jang*, "Single-shot imaging of phase objects fully enclosed by dynamic scattering layers," *Optica*, vol. 13, no. 7, 1245–1253, **2026**. doi:10.1364/OPTICA.593328
2. S. Park, D. Lee, N. Kim, W. Tian, **G. Song**, M. Jang, "Photonics Breakthroughs 2025: Reconstructive spectrometer using double-layer disordered metasurfaces," *IEEE Photonics Journal*, vol. 18, no. 4, 5100307, **2026**. *(Photonics Breakthroughs 2025)* doi:10.1109/JPHOT.2026.3696089
3. T. Yin, **G. Song**, D. Lee, J. Oh, M. Jang, "Impact of system–model mismatch in Fourier ptychographic microscopy," *Advanced Photonics Research*, vol. 6, no. 12, e202500180, **2025**. doi:10.1002/adpr.202500180
4. T. Kwon†, **G. Song**†, Y. Kim, J. Kim, J. C. Ye*, M. Jang*, "Video diffusion posterior sampling for seeing beyond dynamic scattering layers," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 12, 11152–11167, **2025**. doi:10.1109/TPAMI.2025.3598457
5. D. Lee†, **G. Song**†, C. Lee, C. Lee, M. Jang*, "Reconstructive spectrometer using double-layer disordered metasurfaces," *Science Advances*, vol. 11, no. 22, eadv2376, **2025**. doi:10.1126/sciadv.adv2376
6. J. You†, D. Lee†, **G. Song**, C. Lee, M. Jang, "Motion-free high-resolution on-chip microscopy using LED matrix," *Optics Express*, vol. 32, no. 21, 36549–36562, **2024**. doi:10.1364/OE.537127

7. G.-H. Go, D. Lee, J. Oh, **G. Song**, D. Lee, M. Jang, “Meta Shack–Hartmann wavefront sensor with large sampling density and large angular field of view: phase imaging of complex objects,” *Light: Science & Applications*, vol. 13, no. 1, 187, **2024**. doi:10.1038/s41377-024-01528-9
8. A. Shibukawa, R. Higuchi, **G. Song**, H. Mikami, Y. Sudo, M. Jang, “Large-volume focus control at 10 MHz refresh rate via fast line-scanning amplitude-encoded scattering-assisted holography,” *Nature Communications*, vol. 15, no. 1, 2926, **2024**. doi:10.1038/s41467-024-47009-w
9. C. Lee, **G. Song**, H. Kim, J. C. Ye, M. Jang, “Deep learning based on parameterized physical forward model for adaptive holographic imaging with unpaired data,” *Nature Machine Intelligence*, vol. 5, no. 1, 35–45, **2023**. doi:10.1038/s42256-022-00584-3
10. J. Oh, C. Lee, **G. Song**, M. Jang, “Review of endomicroscopic imaging with coherent manipulation of light through an ultrathin probe,” *Journal of Optical Microsystems*, vol. 3, no. 1, 011004, **2023**. *(Invited review)* doi:10.1117/1.JOM.3.1.011004
11. H. Kim, **G. Song**, J. You, C. Lee, M. Jang, “Deep learning for lensless imaging,” *Journal of the Korean Physical Society*, vol. 81, no. 6, 570–579, **2022**. *(Invited review)* doi:10.1007/s40042-022-00412-2

Preprints

1. G.-H. Go, D. Lee, **G. Song**, M. Jang, “Random-mapped intensity optical neural network: all-optical two-layer computing for multimodal optical-field inference,” *arXiv*, 2609.02698, **2026**. doi:10.48550/arXiv.2609.02698

Conference Presentations

1. **[Oral] G. Song**, D. Lee, M. Jang, “Spectrometer using stacked disordered metasurfaces,” *Computational Optical Sensing and Imaging (COSI), Optica Imaging Congress · paper CTh1B.1*, Seattle, WA, USA, **Aug 2025**. — *Student Paper Prize*
2. **[Poster] G. Song**, G. Go, M. Jang, “Meta Shack-Hartmann wavefront sensor with enhanced spatial resolution and acceptance angle for incoherent phase imaging,” *2024 Next Generation Lithography + Patterning Conference (NGL 2024) · poster PS2-05*, Suwon, Korea, **Aug 2024**. — *Best Student Paper Award, 1st Place*
3. J. Oh, H. Ko, **G. Song**, S. Lee, M. Jang, “Deep-tissue optical focusing with phase conjugation based on nonlinear acousto-optic modulation,” *CLEO/Pacific Rim · paper Th1G_2*, Incheon, Korea, **Aug 2024**.
4. **[Poster] G. Song**, D. Lee, C. Lee, C. Lee, M. Jang, “Double-layered disordered metasurface spectrometer,” *Korea International Semiconductor Conference on Manufacturing Technology (KISM 2023)*, Busan, Korea, **Nov 2023**.
5. **[Oral] G. Song**, M. Jang, “Resolution enhancement in ultrasound-modulated optical tomography by detecting high-order diffracted light waves,” *Optics and Photonics Congress (OPC 2021)*, Busan, Korea, **Jul 2021**.
6. **[Oral] G. Song**, D. Lee, M. Jang, “Second-harmonic detection of ultrasound-modulated light,” *32nd Annual Meeting of the Optical Society of Korea (OSK 2021 Winter)*, Korea, **Feb 2021**.

Patents & Inventions

- M. Jang, **G. Song**, C. Lee, D. Lee, “Ultrasensitive spectrometer,” US 12,571,712 B2 (granted Mar 10, 2026); KR 10-3018644 (granted Sep 8, 2026); CN 118936636 A (pending Nov 12, 2024).
- M. Jang, **G. Song**, D. Lee, “Hyperspectral image sensor and system employing the same,” KR 10-2926493 (granted Feb 6, 2026); US 2024/0280406 A1 (pending Aug 22, 2024); CN 118518605 A (pending Aug 20, 2024).
- M. Jang, **G. Song**, “Compact microscope based on random encoding metasurface,” KR 10-2025-0132777 (pending Sep 16, 2025); CN 202610108313.1 (pending Jan 27, 2026); US 19/465,524 (pending Jan 30, 2026).
- M. Jang, J. C. Ye, T. Kwon, **G. Song**, “Method and apparatus for video restoration beyond dynamic scattering media using video diffusion posterior sampling,” KR 10-2026-0031241 (pending Feb 19, 2026); US 19/815,604 (pending Sep 22, 2026).

Inventor since high school — Founded and led Sekwang High School's first invention and patenting club (2012–2014), ran workshops and guided members through filing. Three applications of my own: *Mechanism for contact-lens application* · KR 10-2013-0090667; *Repair tool for mechanical-pencil tips* · KR 20-2014-0009434; *Pen-cap holder cradle* · KR 20-2014-0009014.

Grants & Funded Projects

Lab Startup KAIST 2026 · **selected, ₩35M** — KAIST Startup Support Center 2026

- Team lead. A metasurface-based ultra-compact microscope; KAIST internal R&D program for pre-startup teams, Jul–Nov 2026.

Master's & Ph.D. Adventure Research · **finalist, ₩10M** — KAIST College of Engineering 2021

- Student leader. A portable 3D fluorescence imager built on deep neural networks and scattering media, for miniaturised nucleic-acid diagnostics.

Preliminary Startup Team Support · **₩9M** — KAIST Startup Support Foundation 2021

- Student leader. A portable nucleic-acid diagnostic device based on on-chip hyperspectral imaging.

Research Experience

Sponsored projects of Prof. Mooseok Jang (PI); participating researcher.

Beam shaping with metasurfaces · *LG Electronics* 2025 – 2026

- Fabricated a wide-field-of-view metasurface for an ultra-compact projector.
- Developed an efficient large-scale wave-propagation simulator.

Computational imaging for semiconductor inspection · *Samsung Electronics DS* 2022 – 2025

- Built a fully functional Fourier ptychographic microscope.
- Developed aberration-correction algorithms and alignment protocols for industrial inspection.
- Engineered inverse-design techniques for stacked-metasurface phase maps.

Deep-tissue microscopy by spatiotemporal gating · *National Research Foundation of Korea* 2021 – 2025

- Developed a differential-detection method for high-resolution ultrasound-modulated optical tomography.
- Synchronised ultrasonic and optical pulses for spatiotemporal gating; 2 talks and an M.S. thesis.

Snapshot hyperspectral sensing with 2.5D disordered metastructures · *Samsung Future Technology Center* 2020 – 2025

- Designed a predictive sensor-response model for stacked disordered metasurfaces.
- Built an ultrafast wavelength–space scanner with a supercontinuum laser for real-time hypercube acquisition.
- Demonstrated snapshot hypercube measurement through scattering media; two patent families, now granted in the US and Korea.

Technical Skills

Cleanroom

- *Deposition:* PECVD (SiO_2, SiN_x), Thermal evaporation, E-beam evaporation
- *Lithography:* Photolithography, E-beam lithography, Soft lithography
- *Etching:* ICP-RIE, Wet etching, Lift-off

Optical bench

- *Holography:* Off-axis, Phase-stepping (EOM, AOM), Frequency-modulated (AOM + focused ultrasound), Time-gated (fs laser), Pulse-synchronised (ns laser + pulsed ultrasound)
- *Wavefront shaping:* Digital optical phase conjugation (DMD, SLM)
- *Systems built:* Fast-scanning monochromator, Hyperspectral object projector, Confocal microscope, Fourier ptychographic microscope

Software

- *Programming:* Python (PyTorch, TensorFlow), MATLAB, C/C++
- *Simulation:* FDTD, RCWA, Large-scale wave propagation
- *AI-native development:* Simulators, GUI software for driving lab hardware, Web services

Hardware prototyping

- *Electronics:* PCB design
- *3D modeling & graphics:* Fusion 360, Blender, Maya, Illustrator
- *Fabrication:* FDM 3D printing, SLA/DLP 3D printing, Laser cutting, Prototype building

Last updated Sep 29, 2026